

Turing machines

Exercise sheet 3

Exercise 3.1. Construct Turing machines computing the following functions (with numbers represented in binary): (a) $n \mapsto 2n$; (b) $n \mapsto n + 1$.

Exercise 3.2. Describe Turing machines that convert a natural number: (a) from unary encoding into binary encoding; (b) from binary encoding into unary encoding. (In unary encoding, $n > 0$ is represented by n consecutive 1s, and 0 by ε).

Exercise 3.3. Describe a Turing machine that computes the length of an input word over an arbitrary alphabet $\Sigma \supseteq \{0, 1\}$, writing the answer in binary on the tape.

Exercise 3.4. Describe Turing machines computing the addition function $\langle n, m \rangle \mapsto n + m$, using: (a) unary encoding; (b) binary encoding.

Exercise 3.5. Describe a Turing machine computing the multiplication function $\langle n, m \rangle \mapsto n \cdot m$ using unary encoding.

Exercise 3.6. Prove that the following versions of Turing machines are equivalent to the original in computational power: (a) machines with a tape that is infinite only in one direction; (b) machines with several tapes and several read/write heads; (c) machines that never stay put—every transition shifts them either right or left on the tape.

Exercise 3.7. Describe how a Turing machine could simulate gadgets of modern programming languages: (a) **for** loops; (b) **while** loops; (c) function composition; (d) recursion.

Exercise 3.8. A machine is called a *tape cleaner* if, started in any configuration, it eventually stops in the configuration $q_0\sqcup$ (i.e. it cleans all the content off the tape and stops). Prove that such a machine does not exist.

Exercise 3.9. Describe (in broad strokes) a suitable encoding of Turing machines as binary words $[M] \in \{0, 1\}^*$ and a *universal Turing Machine* U such that, for any machine M computing the function f_M , and any input word w , $f_U(\langle [M], w \rangle) = f_M(w)$.

Exercise 3.10. What can be said about a computation if the exact same configuration appears twice during its execution?

Exercise 3.11. Let $f: \{0, 1\}^* \rightarrow \mathbb{N}$ be the partial function defined as follows: $f(\langle M, x \rangle) = k$ if the Turing machine M on input x visits the blank symbol $\sqcup$ for the last time at step k . If no such step k exists (that is, if M visits $\sqcup$ infinitely often, or never visits $\sqcup$ at all), then $f(\langle M, x \rangle)$ is undefined. Prove that the function f is computable.